



(Economics) (S) (Review Guide)

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Range: Free trade and Trade Protectionism

Note: (This Review Guide) is an UNOFFICIAL source of reference and does NOT include or hint at questions in the actual exam!

Free Trade

Why Do Nations Trade?

Because no country can effectively produce everything it needs, nations trade with one another.

Trade makes it possible for countries:

- To have access to products and services that they are unable to manufacture in their own country.
- Sell extra items to other nations.
- Gain from specialization, which boosts economic development and productivity.

1. Absolute Advantage:

- A nation has an absolute advantage if it can manufacture a good with less labour and capital than another nation.

For instance, in this case, Japan has an absolute advantage in the camera industry but Sweden has an absolute advantage in making cars.

If each country has an absolute advantage in one good and specializes in that good, then both countries can gain from trade.

specialization:

Sweden only produces cars and Japan only produces cameras)

Country	Cars	AND	Cameras
Sweden	<u>100</u>		0
Japan	0		<u>100</u>
Total	<u>100</u>		<u>100</u>

after trade (suppose 1 car exchange for 1 camera):

Country	Cars	AND	Cameras
Sweden	<u>70</u>		<u>30</u>
Japan	<u>30</u>		<u>70</u>
Total	<u>100</u>		<u>100</u>

→ Specialization and trade make both countries have more cars and cameras than they did before.

Japan makes more cameras, and Sweden makes more cars, so by trading they can both benefit from each other while making a profit.

2 Comparative Advantage:

- A country has a comparative advantage when it can produce a good at a lower opportunity cost (what it gives up to produce that good) than another country.
- Example: New Zealand may produce fruit more efficiently than industrial goods, so it should focus on fruit and trade for industrial goods.

Country	Fruit (tons)/per 100 labor hour	OR	Industrial goods (tons)/per 100 labor hour
China	35		21
New Zealand	30		6
Total	65		27

- Opportunity cost of one ton of fruit:
 - China 0.6 industrial goods
 - New Zealand 0.2 industrial goods
- Opportunity cost of one ton of industrial goods:
 - China 1.67 fruit
 - New Zealand 5 fruit
- China has a **comparative advantage** in producing industrial goods
- New Zealand has a **comparative advantage** in producing fruits

Opportunity cost of one ton of fruit: Industrial goods / Fruit

China: $21 / 35 = 0.6$

So, for every fruit, China sacrifices 0.6 industrial good that could've been produced.

New Zealand $6 / 30 = 0.2$

So, for every fruit, New Zealand only sacrifices 0.2 industrial goods that could've been produced.

China produces more fruit than New Zealand but their opportunity cost of one ton is higher.

Therefore, trading industrial goods for fruit will be beneficial for China.

Opportunity cost of one ton of Industrial goods: Fruit / Industrial goods

China: $35 / 21 = 1.67$

For every industrial good produced China sacrifices 1.67 fruits that could've been produced.

New Zealand: $30 / 6 = 5$

For every industrial good produced, New Zealand sacrifices 5 fruits that could've been produced.

New Zealand has a higher opportunity cost when producing industrial goods, so trading fruit for Industrial goods will benefit them.

How Trade Benefits Nations:

- Increased Efficiency: Countries specialize in what they do best, leading to higher global output.
- Lower Prices: Consumers benefit from cheaper imported goods (e.g., electronics, clothing).
- Greater Variety: Trade provides access to products not available domestically.
- Economic Growth: Exporting industries expand, creating jobs and boosting GDP.

Potential Downsides of Free Trade:

- Job Losses: Industries that cannot compete may shrink (e.g., U.S. manufacturing moving overseas).
- Dependency: Over-reliance on imports can be risky (e.g., shortages during global crises).
- Environmental & Labour Concerns: Some countries may exploit workers or natural resources to cut costs.

World Price and comparative advantage

PW = the world price of a good, the price that prevails in world markets.

PD = domestic price without trade

If $PD < PW$,

Then this country has comparative advantage in the good under free trade, US exports the good

If $PD > PW$,

Then this country does not have comparative advantage under free trade, US imports the good

Take this example, where $PD > PW$ in the EV cars industry

US Imports Chinese EV cars

Without trade,

$$CS = A$$

$$PS = B + C$$

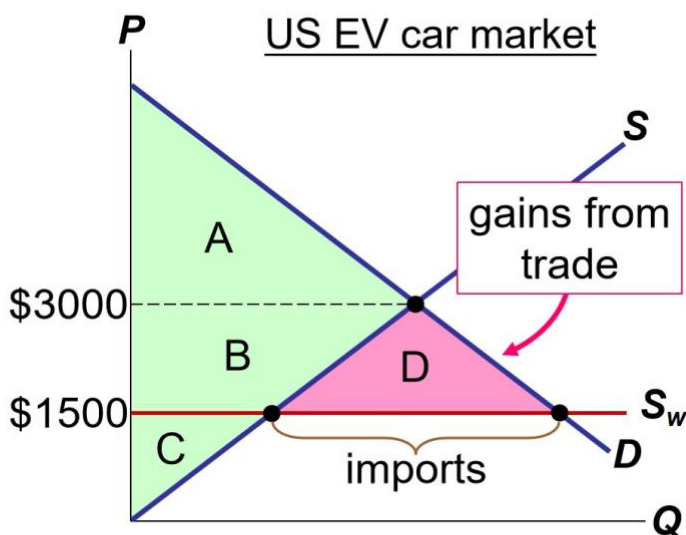
$$\begin{aligned} \text{Total surplus} \\ = A + B + C \end{aligned}$$

With trade,

$$CS = A + B + D$$

$$PS = C$$

$$\begin{aligned} \text{Total surplus} \\ = A + B + C + \boxed{D} \end{aligned}$$



In this case, the domestic price is higher, so therefore imports with World Price (cheaper) are encouraged. This causes the price to decrease and the total surplus to increase, benefiting the US with more gains, as shown in the graph.

Trade Protectionism

What Is Protectionism?

Protectionism refers to government policies that restrict free trade to shield domestic industries from foreign competition.

Some methods include:

1. Tariffs (Taxes on Imports)

- Example: The U.S. imposes a 247.5% tariff on Chinese electric vehicles (EVs), making them more expensive.

- **Pros:**

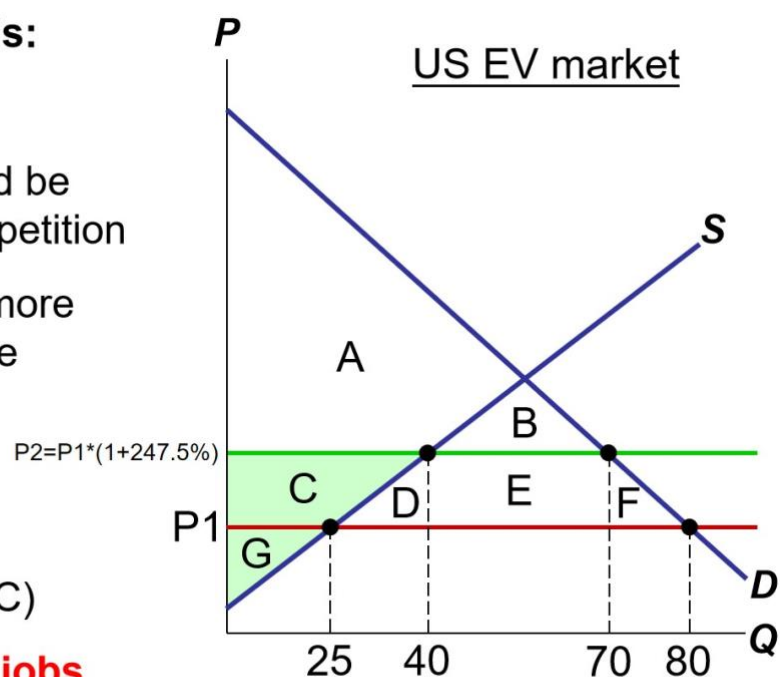
- US producers will have less foreign competition, because products from China will be too expensive for the consumers, and therefore more of their own product will be bought by the Americans. More business causes, more production, revenue, employment etc...

Stakeholder analysis:

2) US producers:

- Higher prices could be charged, less competition
- More production, more market share, more revenue
- More employment
- Producer surplus increased (G → GC)

Better off, more jobs created and maintained



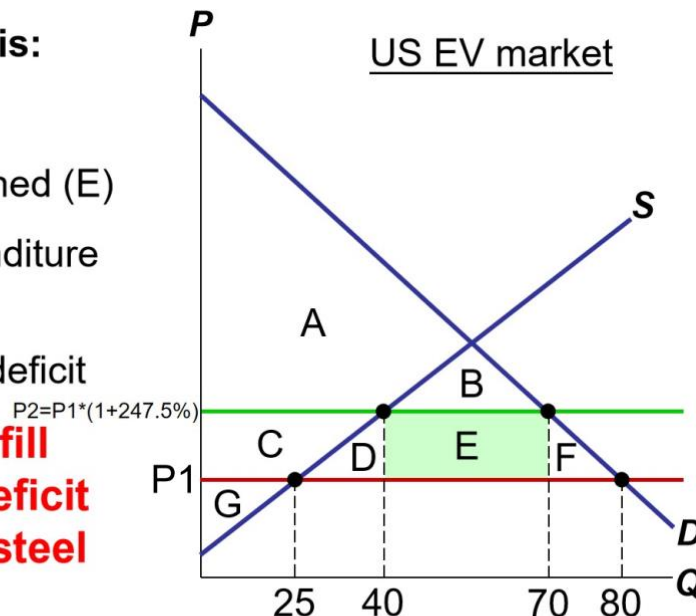
- The US government also benefits, because the extra money producers pay due to tariffs becomes government revenue which can later be used to build infrastructure, fund public services etc...

Stakeholder analysis:

3) US government:

- Tariff revenue gained (E)
- Less import expenditure paid
- Narrow the trade deficit

**Better off, fulfill
budget/trade deficit
and subsidize steel
industry**



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-Cons:

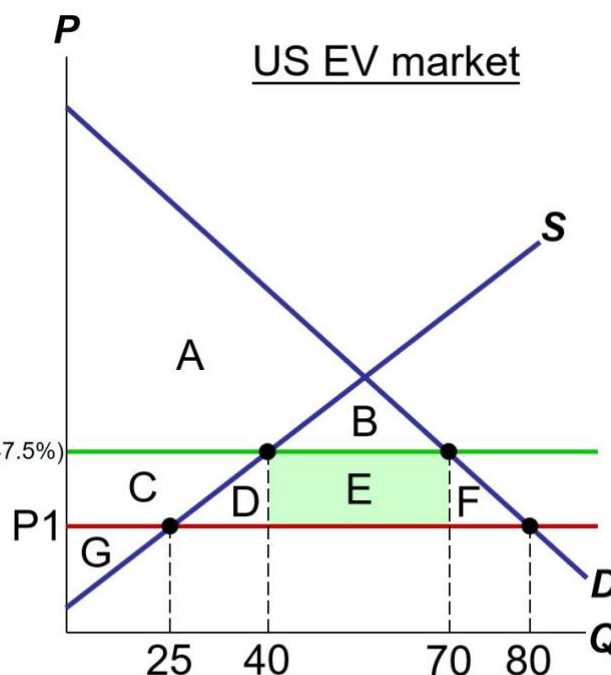
- Chinese exporters will lose a lot of the American market because their products will become more expensive for American buyers. This causes less exports and therefore less revenue, production and employment.

Stakeholder analysis:

4) Chinese exporters:

- Less export (55 → 30), less export revenue
- Lose part of US market, may find new market to export
- May find ways to retaliate

Worse off



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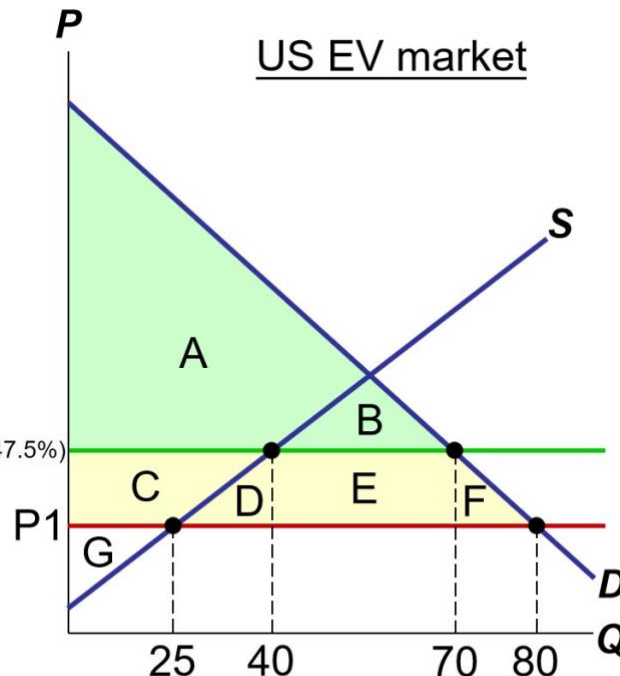
-US consumers will have to pay higher prices because Chinese products that are cheaper than domestic products will have tariffs on them. This causes demand and consumer surplus to decrease.

Stakeholder analysis:

1) US consumers:

- Higher prices
- Less quantity demanded, fewer choices
- Consumer surplus decreased (ABCDEF → AB)

Worse off, especially for those manufacturing firms



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2. Import Quotas (Limits on Quantity of imported goods)

- Example: The EU restricts the number of U.S. steel imports.
- Pros: Helps local producers by reducing competition and protects local businesses.
- Cons: Shortages, higher prices, and no extra revenues for government

3. Subsidies (Government Payments to Domestic Firms)

- Example: India subsidizes sugar exports to compete globally.
- Pros: Supports struggling industries, encourages domestic production
- Cons: Can lead to overproduction and trade disputes (WTO often bans unfair subsidies).

4. Non-Tariff Barriers (Regulations, Standards, Bureaucracy)

- Example: The EU bans hormone-treated U.S. beef, citing health concerns.
- Pros: Protects consumers from unsafe products.
- Cons: Can be used as disguised protectionism.

Arguments for Protectionism:

- Protecting Jobs (e.g. tariffs on steel save U.S. factory jobs).
- National Security (e.g. limiting reliance on foreign weapons or food).
- Infant Industry Protection (helping new industries grow before competing globally).
- Preventing Unfair Competition (e.g. China's state subsidies for EVs).

Arguments Against Protectionism:

- Higher Consumer Prices (tariffs make imports costlier).
- Trade Wars (tariffs hurt exporters and might encourage them to implement retaliatory tariffs, like China did to the US).
- Inefficiency (protected industries may become lazy without competition).
- Deadweight Loss (economic waste from reduced trade).